Scheme/ Answer Key for Valuation

Scheme of evaluation (marks in brackets) and answers of problems/key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.TECHS2 (R) EXAMINATION, JULY 2021 (2019 SCHEME)

Course Code: CYT100

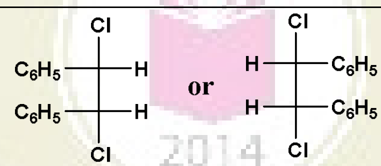
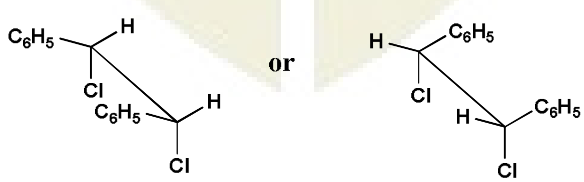
Course Name: ENGINEERING CHEMISTRY

(**2019-Scheme**)

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

1	Definition (1.5) Reason (1.5)	(3)
2	On Charging at anode $LiCoO_2 \rightarrow Li_{1-x}CoO_2 + xLi^+ + xe$ On 100% charging x=1 $LiCoO_2 \rightarrow CoO_2 + Li^+ + e$ CoO_2 is an explosive material which on decomposition gives O_2 and Co_2O_3 Any one equation (1) Reason (2)	(3)
3	Correct reason by explaining the cumulative effect of electronegativity (3)	(3)
4	Electronic transitions are always accompanied by vibrational and rotational transitions. (2) Diagram (1)	(3)
5	Any 3 applications -3 marks	(3)
6	Retention time - 2 RPA - 1	(3)
7	 Fischer projection of *meso*-stilbene dichloride  Saw-Horse structure of *meso*-stilbene dichloride Fischer representation of *meso* form – **2** marks Saw-Horse structure – **1** mark	(3)

8		Any one method (2) equation (1)	(3)
9		NH_4Cl-NH_4OH buffer to maintain pH around 10 (1mark) Only at this pH indicator EBT can form complex (2 marks)	(3)
10		Graph (1) explanation (2)	(3)
PART B			
Answer one full question from each module, each question carries 14 marks			
Module-I			
11	a)	Derivation (5) Applying on Daniel cell (3)	(8)
	b)	Procedure (2) Equations (2) two applications (2)	(6)
12	a)	Oxygen electrode potential at pH=0 is 1.23 V and pH=14 is 0.41V $E^0_{Fe2+/Fe}$ = -0.44 V Corrosion cell potential in acidic oxygen rich condition is 1.67V , which is larger than corrosion cell potential in alkaline medium 0.85V Electrode reaction and Nernst equation for oxygen electrode – (2) mark, Variation of oxygen electrode potential with pH – (2), Calculation of E^0cell at acidic and alkaline condition – (2), Reason for rapid rusting acidic condition – (2)	(8)
	b)	Ecell= (E^0_G-0.2422)-0.0595pH E=A+B x PH A- y-intercept B-slope 212=A+Bx4 --(1) -30 =A+Bx9.2 --(2) On solving B = -242/5.2= -46.53 A= 398.12V pH of test solution = (120-398.12)/-46.53 = 5.97 E^0_G=0.39812+0.2422= 0.6403 V Equations (2) Calibration (slope & intercept)-(2) pH-(1) E^0_G-(1)	(6)
Module-II			
13	a)	Energy level diagram with explanations 2 marks each	(8)
	b)	Number signals – 1 mark, explanation of neighbouring protons - 1 spin – spin splitting - 4	(6)
14	a)	Principle (3) Equation for energy $E = (V + \frac{1}{2})h\nu_0$, $E_0 = \frac{1}{2}h\nu_0$ (3) PE diagram (2)	(8)

	b)	Ketone –acetone only one peak (3), propanal three peaks (3) (chemical shifts) Or Any other proper explanation related to the aldehydic proton in the ^{1}H NMR spectrum (singlet around 9.3 ppm)	(6)
Module-III			
15	a)	Principle (2) procedure including solvent polarity (6) TLC monitoring (2)	(10)
	b)	DTG graph (4) Plot of DTA profile can also be considered for giving 3 marks.	(4)
16	a)	Hydrolysis-Hydrothermal (3) Solgel (3) Reduction Chemical (2) Electro (2)	(10)
	b)	Procedure (4)	(4)
Module-IV			
17	a)	Two asymmetric carbon atoms- so 2^2= 4 forms – 1 mark Two enantiomeric pairs , 4 fischer formula 1.5 x 4 All are optically active – 1 marks	(8)
	b)	Definition (1), construction: figure (1) explanation (2), working (2)	(6)
18	a)	Definition (2), 4 types with examples (2 x4)	(10)
	b)	Structure of 3 monomers (3) any two applications (1)	(4)
Module-V			
19	a)	Trickling filter with diagram - 5 UASB with diagram -5	(10)
	b)	Procedure -4	(4)
20	a)	Definition (2) method with diagram (2+2) Two advantages (2)	(8)
	b)	Process (2) equations (2) regeneration (2)	(6)
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